3.4 FinTech Analytics

Complete Notes: Growth Analytics • Unit Economics • Alternative-Data Credit Scoring • Personalisation • Four Types of Analytics

FinTech = Financial Technology. FinTech means using technology, software, data, analytics, automation and related digital systems to provide financial services.

Examples: digital payments, digital lending, BNPL, investment platforms, digital banking, insurance technology, fraud detection, credit scoring and wealth management.

1. Growth Analytics

Growth Analytics uses data to understand how a FinTech product is growing and what is causing that growth.

Questions: Are users increasing? Are users actually using the application? Where are users dropping out? Which activities generate revenue? Which acquisition channel brings valuable customers?

2. DAU and MAU

DAU = Daily Active Users — the number of unique users who actively use an application on a particular day.

Example: If 50,000 unique users actively use the application on Monday, DAU = 50,000.

MAU = Monthly Active Users — the number of unique users who actively use an application during a month.

Example: If 5,00,000 unique users use the app during August, MAU = 5,00,000.

DAU/MAU Ratio is an engagement metric. Formula: DAU/MAU × 100.

Example: DAU = 1,00,000 and MAU = 5,00,000 → (1,00,000 / 5,00,000) × 100 = **20%**.

A higher ratio generally indicates more frequent usage. The appropriate benchmark depends on the type of FinTech product; it is not a universal industry standard.

3. Funnel Conversion

A **conversion funnel** represents the customer journey from first interaction to a desired business action.

Typical FinTech funnel: Advertisement → App Install → Registration → KYC → Bank Linking → First Transaction → Repeat Transaction.

KYC = Know Your Customer.

Stage	Users
Advertisement Clicks	1,00,000
App Installs	50,000
Registrations	40,000
KYC Completed	30,000
Bank Account Linked	25,000
First Transaction	15,000
Repeat Transaction	10,000

Install → Registration: 40,000 / 50,000 × 100 = 80%.

Registration → KYC: 30,000 / 40,000 × 100 = 75%.

KYC → First Transaction: 15,000 / 30,000 × 100 = 50%.

The analyst can identify exactly where users are dropping out and investigate registration difficulty, KYC delays, bank-linking failures, confusing interfaces or other friction.

4. Unit Economics

Unit Economics studies the economics of one customer, transaction or business unit. The core question is: How much does the company spend to acquire a customer, and how much value does that customer generate?

CAC = Customer Acquisition Cost. Formula: CAC = Total Marketing and Sales Cost / Number of New Customers.

Example: ■10,00,000 spent to acquire 10,000 customers → CAC = **■100/customer**.

LTV = Lifetime Value. It estimates the value a customer generates during the relationship with the company.

Simplified formula: LTV = Average Revenue per Customer × Customer Lifetime.

Example: Monthly contribution = ■200 and lifetime = 24 months → LTV = ■4,800.

Important: In real FinTech analysis, revenue is not automatically profit. Analysts may use contribution margin after considering transaction costs, servicing costs, incentives, credit losses and other relevant costs.

5. CAC vs LTV

Example: CAC = ■100 and LTV = ■4,800. The simple LTV:CAC ratio is 48:1. The business should compare acquisition cost with the economically meaningful customer value, not revenue alone.

6. CAC by Acquisition Channel

FinTechs may acquire customers through Google Ads, Instagram, YouTube, referrals, affiliates and organic search. The key question is not only which channel acquires the most users, but which channel acquires valuable users at an acceptable cost.

Channel	Cost	Customers	CAC
Google Ads	■5L	5,000	■100
Instagram	■4L	2,000	■200
Referral	■2L	4,000	■50
Affiliate	■3L	1,500	■200

For example, referral customers with CAC ■50 and LTV ■3,000 can be more economically attractive than Instagram customers with CAC ■200 and LTV ■1,000.

7. Alternative-Data Credit Scoring

Alternative Data means relevant data beyond traditional credit-bureau information that may provide additional signals for assessing creditworthiness.

Thin-file borrower: a borrower with insufficient traditional credit history to confidently assess their creditworthiness.

A small shop owner may have limited loan history but may have meaningful business activity. With lawful access, consent where required and responsible data practices, FinTech lenders can evaluate additional signals such as UPI cash flows, GST-related information and relevant app/financial behaviour.

8. UPI Cash Flow

UPI = Unified Payments Interface. UPI is India's real-time digital payment system.

Potentially useful signals include transaction volume, transaction frequency, cash-flow patterns, business activity and seasonality.

Important: cash flow is not automatically equal to income or profit. It needs careful interpretation.

9. GST Returns

GST = Goods and Services Tax. GST-related records may provide information about reported business activity such as sales, purchase patterns, growth, seasonality and filing behaviour, subject to lawful access and applicable rules.

10. App Data

Relevant app/financial behaviour can include usage frequency, transaction patterns, repayment behaviour, product usage and application interactions. A lender should not simply collect everything available. Responsible analytics requires privacy, consent where required, data minimisation, security, fairness, explainability and regulatory compliance.

11. BNPL and Small-Business Lending

BNPL = Buy Now, Pay Later. It allows eligible customers to defer payment or repay according to an approved schedule.

For small-business lending, alternative signals can potentially help understand business cash flow and stability when traditional credit information is limited. Models should be validated and monitored for bias, errors and changing economic conditions.

12. Personalisation

Personalisation means showing relevant products, offers, messages or experiences to different customers based on appropriate customer characteristics and behaviour.

Example: A customer who frequently uses merchant payments may receive a relevant merchant QR payment offer rather than a generic offer.

13. Real-Time Behaviour

Suppose a customer opens an app, searches for a loan, checks eligibility, views EMI options and leaves without applying. A FinTech system may identify this journey and show a relevant reminder or information, subject to customer preferences and applicable rules.

EMI = Equated Monthly Instalment.

14. Four Types of Analytics

1. Descriptive Analytics — What happened? It summarises historical/current data. Example: MAU increased by 10%.

2. Diagnostic Analytics — Why did it happen? It investigates causes. Example: new users mainly perform utility bill payments.

3. Predictive Analytics — What is likely to happen? It uses historical data and statistical/machine-learning models to estimate future outcomes such as churn, conversion, response or repayment risk.

4. Prescriptive Analytics — What should we do? It recommends or evaluates actions, such as changing onboarding to encourage higher-value merchant QR payments.

15. Complete Real-Life Example

Scenario: A payments FinTech notices that MAU is growing 10%, but revenue is flat.

Step 1 — Descriptive: The dashboard reports MAU +10%.

Step 2 — Diagnostic: Drill-down shows that new users are mostly doing low-margin utility bill payments instead of higher-value merchant payments.

Step 3 — Predictive: Historical data indicates that users who make merchant QR payments early in their journey are more likely to become valuable active users. A model can estimate which new users may respond to merchant-payment onboarding.

Step 4 — Prescriptive: Change onboarding to highlight merchant QR payments, provide education and, where commercially and legally appropriate, an introductory incentive.

This one insight cycle uses all four analytics types: **What happened?** → **Why?** → **What may happen?** → **What should we do?**

16. Analytics Cycle

FINTECH DATA → DESCRIPTIVE → DIAGNOSTIC → PREDICTIVE → PRESCRIPTIVE → BUSINESS ACTION → NEW DATA → ANALYTICS

Think of it as a continuous **Data** → **Insight** → **Action** → **Result** → **Data** cycle.

17. What a Business Analyst Does in FinTech

Excel / SQL: analyse customer transactions, activity, revenue, marketing costs and channel performance.

Python: NumPy for numerical calculations, Pandas for cleaning/analysis, Matplotlib and Seaborn for visualisation, and Scikit-learn for predictive modelling.

Power BI: dashboards for DAU, MAU, DAU/MAU, conversion, CAC, LTV, revenue, transaction volume, channel performance and customer segments.

Business decision: translate analysis into a clear recommendation for management. A Business Analyst is not only a person who creates charts; the goal is to solve business problems using evidence.

18. Important Full Forms

Term	Full Form	Meaning
FinTech	Financial Technology	Technology used for financial services
DAU	Daily Active Users	Active users in a day
MAU	Monthly Active Users	Active users in a month
CAC	Customer Acquisition Cost	Cost to acquire one customer
LTV	Lifetime Value	Value generated by a customer over their relationship
UPI	Unified Payments Interface	Real-time digital payment system in India
GST	Goods and Services Tax	India's indirect tax system
BNPL	Buy Now, Pay Later	Deferred-payment/short-term credit product
KYC	Know Your Customer	Customer identity verification process
QR	Quick Response	Machine-readable code commonly used for payments
BA	Business Analyst	Professional who analyses business problems/data
KPI	Key Performance Indicator	Metric used to measure performance
ROI	Return on Investment	Return generated relative to investment

19. Quick Memory Trick

FinTech Analytics = Growth → Unit Economics → Credit Risk → Personalisation

Four analytics = Descriptive → Diagnostic → Predictive → Prescriptive

Memory questions: What happened? → Why did it happen? → What is likely to happen? → What should we do?

Note: This document is designed as educational material. Real FinTech credit, customer-data and personalisation systems must follow applicable laws, regulations, consent requirements, privacy/security standards and responsible-lending practices.